(a) For hydrodynamically rough boundary, the friction coefficient

(i) is constant

(ii) varies inversely with Reynolds number

(iii) is function of Reynolds number and relative roughness

(iv) is dependent on relative roughness only

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(iv) Manning's coefficient

(d) Reynolds number may be defined as the ratio of

(i) viscous forces to inertial forces

(ii) elastic forces to pressure forces

(iii) inertial forces to viscous forces

(iv) gravity forces to inertial forces

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(3) layer is defined as the distance from the wall to a point

(i) where the velocity is 99% less than the asymptotic limit

(ii) where the velocity ceases to be laminar

(iii) where the velocity is within 90% of the asymptotic limit

(iv) where the velocity is 99% of its asymptotic limit

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(j) Bernoulli equation is applicable between any two points

(i) in any rotational flow of an incompressible fluid

(ii) in any type of irrotational flow of a fluid

(iii) in steady rotational flow of an incompressible fluid

(iv) in steady irrotational flow of an incompressible fluid

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(b) A most efficient trapezoidal section is required to give a maximum discharge of 21·5 m^3/s of water. The slope of the channel bottom is 1 in 2500. Taking $C = 70\ m^{1/2}/s$ in Chezy's equation, determine the dimensions of the channel. Also determine the value of Manning's n, taking the value of velocity of flow as obtained for the channel by Chezy's equation. 6

(iii) convective acceleration at B

(iv) flow per unit width across AB. 8

5. Show that the gradually varied flow equation for flow in a rectangular channel of variable width B may be expressed as

$$\frac{dy}{dx} = \frac{S_0 - S_f - \frac{Q^2 y}{g A^3}\frac{dB}{dx}}{1 - \frac{Q^2 B}{g A^3}}$$

14

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8. (a) What do you understand by *(i)* steady and unsteady flow; *(ii)* uniform and non-uniform flow in the case of channels? 7

(b) A rectangular channel which is laid on a bottom slope of 0·0064 is to carry 20 m^3/s of water. Determine the width of the channel when the flow is in critical condition. Take Manning's $n = 0{\cdot}015$. 7

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